

1 Customization of Applications

1.1 Overview

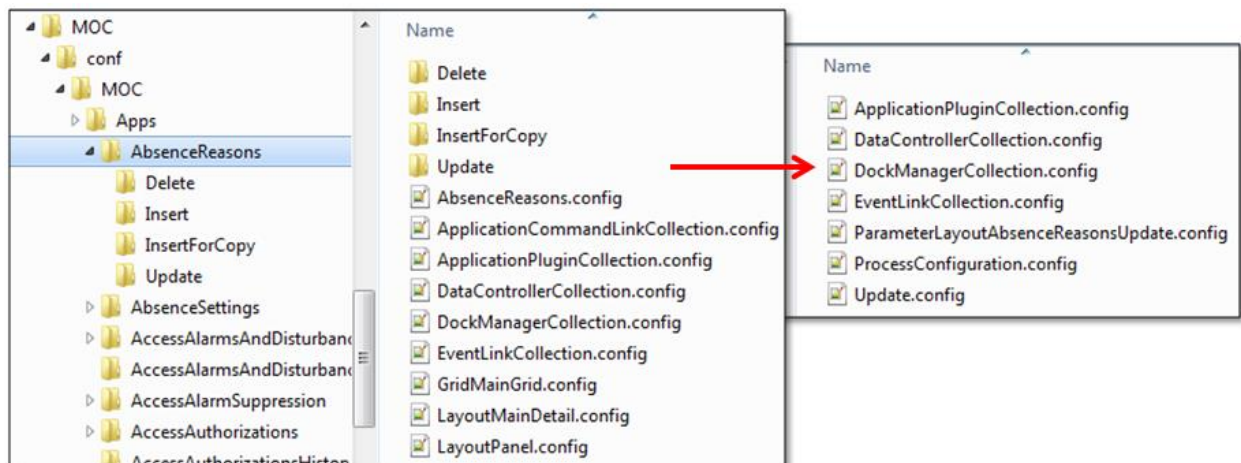
The MOC provides two types of applications:

- Main applications to present data
- Editing applications to edit data

There is a directory with the clear/unambiguous (English) application name for each main application. The application directory includes all files required to customize an application. The specific files and the number of files are different for each application. This largely depends on the number of detail applications (tables, charts etc.) included in an application.

Editing applications are always assigned to a main application. The files for the customization of detail applications are therefore managed in sub folders of a main application. In addition to the customization files of the main application, the application directory contains an additional directory including the files of the detail application.

Example: The files of the application *Absence reasons* are stored in the folder of the applications with the name of the relevant application ID, i.e. in this example "AbsenceReasons". The editing dialogs required to edit absence reasons are stored in the sub folders "delete", "insert" and "update".



1.2 Customization of main applications

Applications are the central elements in order to access the variety of functions of the MES Operation Center. All applications have the same structure, i.e. they contain

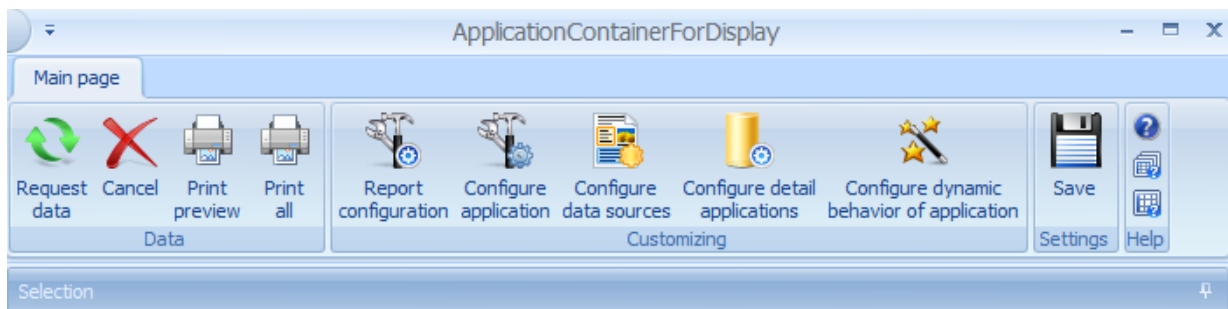
- the main application, i.e. the frame of the application (also called "Application Container")

- a toolbar with preset buttons to activate standard functions (Request data, Print preview, Save configuration, Help, etc.) and a deliberate number of buttons added via customization.
- a selection panel to enter selection or filter criteria with a configurable number of input fields
- data sources (DataControllers) to supply detail applications with (web service) data,
- detail applications embedded in the application via customization, and
- a deliberate number of report configurations to activate the defined reports.

The different components of an application and their customization options are described in detail in the sections below.

1.2.1 Creating an application

You use the menu item *MES Development Suite – New application* to create a new application. A new application opens that can be edited. Save the application to persist the changes.



You cannot create applications that include editing dialogs using this procedure. You then require a specific application generator, as described in section "1.3.1 Application generator".

1.2.2 Customization of applications

You can use the button *Configure application* in the toolbar to edit specific properties of an application. You can edit the following values.

ID

Clear ID of the application that can be used to identify the relevant customization file, for example.

Caption

Text displayed in the title bar of the application. Specify a language key ("lkxxx") for this text so that the caption is translated.

Help file and Help index

Name of file where help subjects for this detail application are listed and/or description of the entry point within this file.

Update rate

Time in seconds after which this data source will automatically and regularly request data. This is the default value for the entire application. This value is used for all detail applications containing the value 0 as update rate.

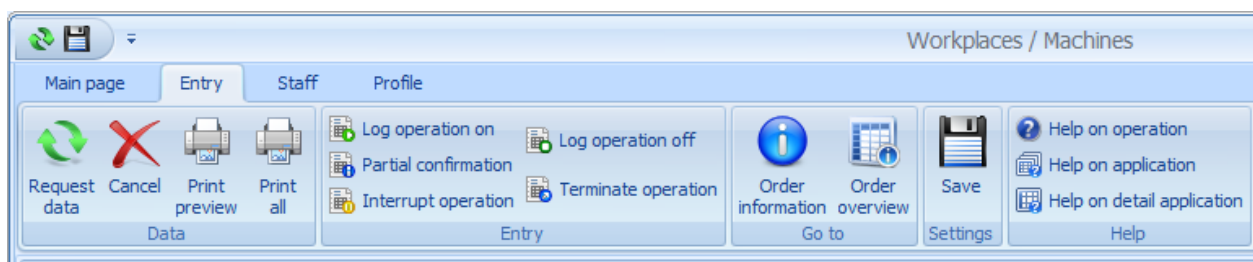
If this value is 0, data is not automatically updated.

Version

Application version

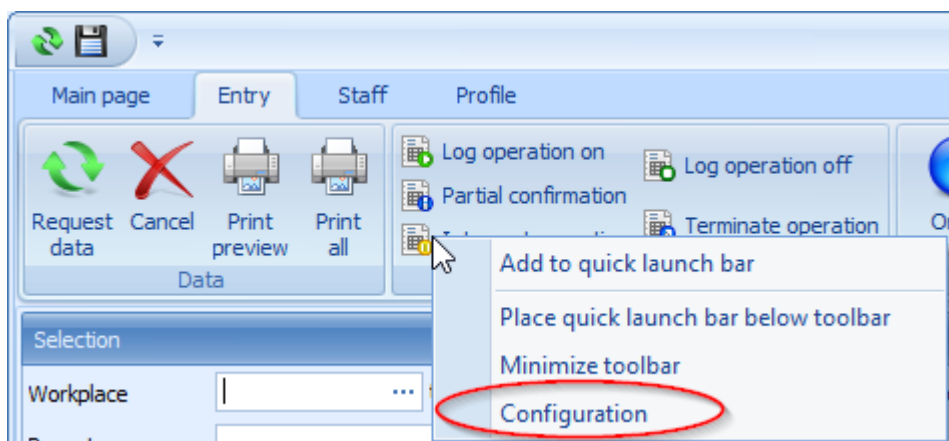
1.2.3 Toolbar

The toolbar provides a quick access to context-related functions. The toolbar can consist of several tabs which can include several categories each.



The categories "Data" and "Help" are given on each tab in each application. The "Data" category includes the buttons *Data request*, *Cancel*, *Print preview* and *Save*. The categories "Data" and "Help" and the buttons contained cannot be changed.

If the MES Development Suite is activated, another non-adjustable category, *Customizing*, is shown. This category includes all important functions to change the application (please also refer to 1.2.4 et sqq.).



The *Configuration* menu item in the context menu of the toolbar calls the dialog to edit the functions. Here, you can create new buttons and edit existing buttons that can be changed.

The following customization options are available:

ID	Clear ID of the ApplicationCommandLink that is called
Command	Deliberate identification of the ApplicationCommandLink
Function	Function that you want to execute. Possible values are e.g. callScript, openApplicationContainerForEdit
Parameter	Parameters passed to the function
Authorization key	Is used to authorize this function via function authorizations.
Title	Language key for display
Category	ID of the category (is a language key). If the category does not exist, it is created.
Ribbon page	ID of menu bar tab (is a language key). If the tab does not exist, it is created.
Comment	Comment
Keyboard shortcut	Keyboard shortcut Valid keyboard shortcuts : CTRL+A to CTRL+Z CTRL+F1 to CTRL+F12 SHIFT+F1 to SHIFT+F12
Image (small)	Image used to present the function in the toolbar
Image (large)	Image for presentation. If a large image is available, it will be used; otherwise the small image is used.

There is no customization file for default categories that already exist. Additional functions are saved in the file ApplicationCommandLinks.config. This file is only created if required (if functions are added and the application is then saved).

Important functions

Link to a main application

ID = application ID

Command = application ID

Function = openAppContainer

Parameter = id=<application ID>

The below parameters can be transferred additionally (added at the end, separated by blanks):



The parameters are case sensitive!

- autorequest=true/false → if true, data is automatically requested after the application has been opened
- windowOpenMode=MultiWindow/SingleWindow → if MultiWindow, a new instance of the application is opened every time it is linked; if SingleWindow the same instance of the application is always opened. MultiWindow is always set by default to link one application to another application. If SingleWindow is set explicitly via parameter transfer, all other transferred values are also reset automatically. Data is only requested automatically if autorequest=true is passed.
- Modal=true/false → if true, the application is opened modally
- selectedprofile=<Profile> → if a defined profile is transferred, it is automatically preassigned when the application is opened.

If the name of the selected profile includes blank characters, the profile name has to be enclosed by "inverted commas".

You can also transfer values from the application (also applies for opening of editing applications):

- zielacronym=DC (the relevant value of the same acronym from DatenController is transferred to the target acronym)
- zielacronym=DC:quellacronym (the relevant value of the source acronym from DatenController is transferred to the target acronym)
- zielacronym=SP (the relevant value of the same acronym from SelectionPanel is transferred to the target acronym)
- zielacronym=SP:quellacronym (the relevant value of the source acronym from the SelectionPanel is transferred to the target acronym)

Opening an editing application

ID = ID of the editing application

Command = ID of the editing application

Function = openApplicationContainerForEdit

Parameter = id=<ID of the editing application>

Calling a script

ID = Script ID

Command = Script ID

Function = callScript

Parameter = id=<script ID>

1.2.4 Selection panel

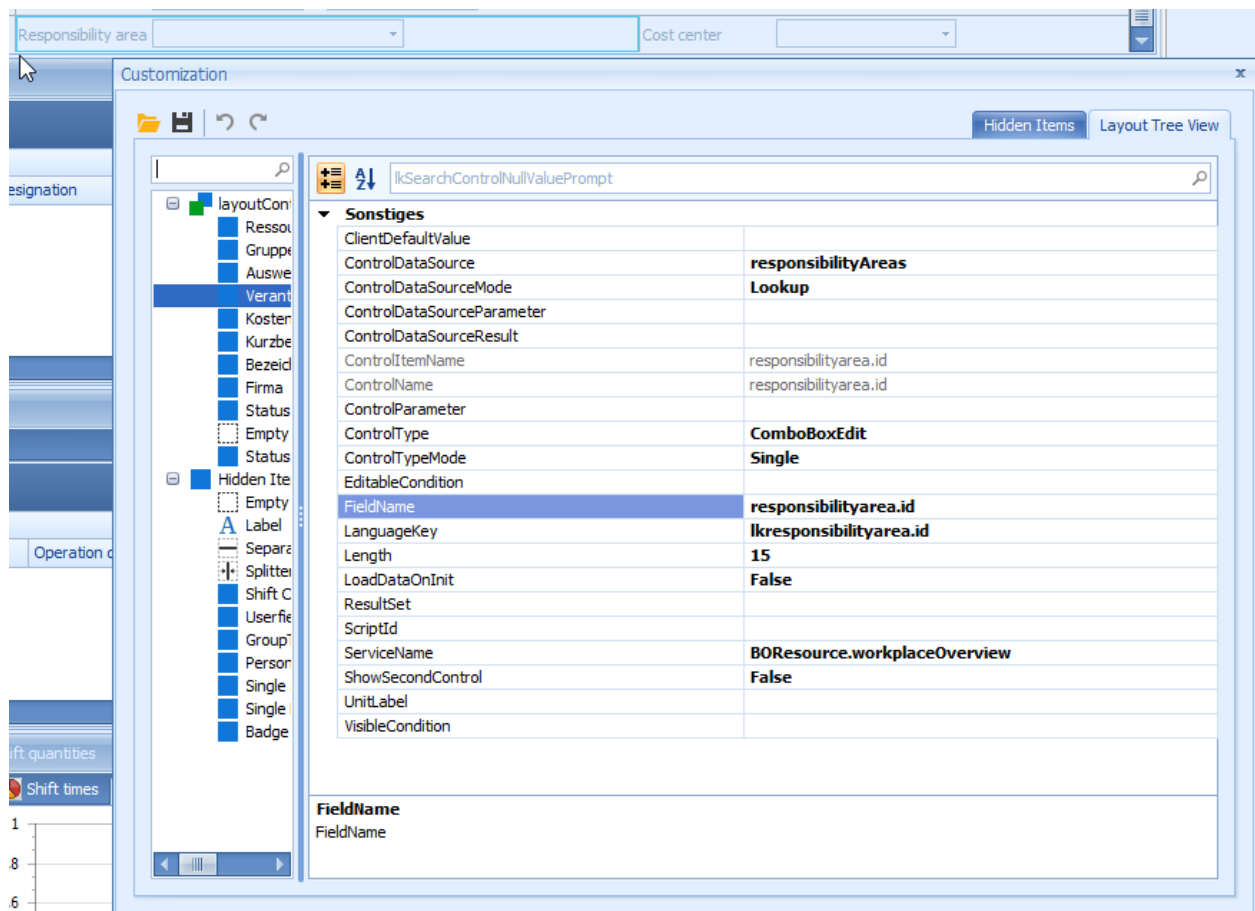
The selection panel contains input fields (controls) to enter filter criteria. To use input fields as filter criteria, the following steps are required.

- Adding of input fields to the selection dialog and
- assigning input fields as parameters for the data sources (this step is described in section 1.2.5)

Activating the editing mode

If the MES Development Suite is activated, right-click the empty space of the selection panel or the label of a control to open the context menu. Use the context menu to add new controls, delete existing controls or to activate the mode to edit layouts and controls (*Customize layout*).

You can use the editing mode to adjust the layout and to specifically edit specific controls in the "Customization" dialog.



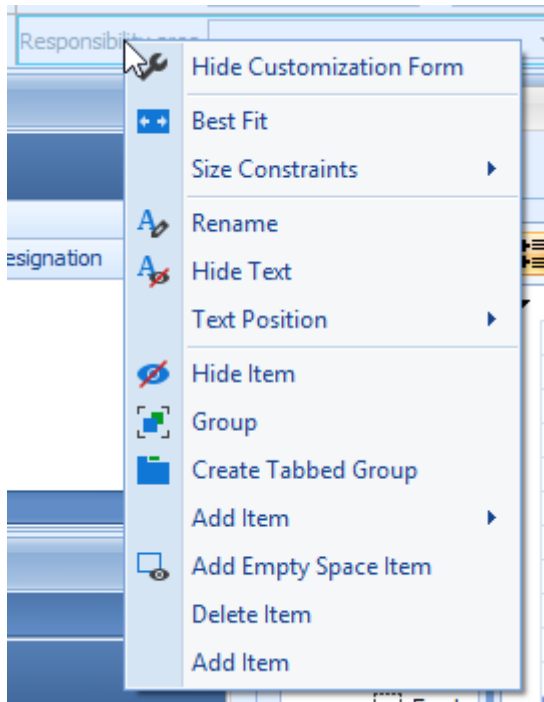
In addition, you can move specific control types to the dialog using drag and drop in editing mode. These can be

- UserField-Control: a control to present user fields (see section "User fields").
- Shift Control: a control to enter shift times
- Person selection: a control to configure person selections in a very flexible manner.

Changing the layout

In the editing mode, you can change the layout of a selection panel. Click the controls and move them using drag & drop. Please note that the layout function will attempt to arrange the controls in columns in order to achieve a very uniform presentation.

In the editing mode, the context menu includes additional functions to edit controls.



To access this context menu, click *Customize layout* in the context menu of the control in the application, and in the *Customization* window go to tab *Layout Tree View* and right-click an item.

Tips for the layout:

- Use *Create EmptySpaceItem* to create forced spaces between controls.
- You can use the values in *Size constraints* to fix height and/or width of controls. After editing, we recommend to fix the values.
- Use *Text Position* to place the label text and *Hide Text* to hide or show the label text.
- Use the item *Group* to combine several fields to an area or *Create Tab Group* to distribute fields to tabs. It is best to use the dialog *Customization* to this end. Here, you can select the controls that you want to group in the tree view and group the controls using the entry of the context menu.

Editing input fields

Controls usually have the type "mpdvEdit". You can add the controls of this type to a layout using *Add item* and delete the controls using *Remove item*. If the editing mode is activated, click a control to show its properties on the right hand side of the dialog *Customization*.

The detailed properties of the controls are described in the "Input and Output Fields" section. Please find some notes on particularly important properties in the context of the selection panel below:

- The "FieldName" is used to map the control value on a service parameter on the one hand, but on the other hand, it may also be used to adopt existing customization settings from other controls automatically, if the field name corresponds to a property and/or a service parameter. It is often enough to select a service parameter to get a completely defined control. The selection list of the service parameters is specified via the contents of the "ServiceName" property - only those service parameters are displayed that are known in the service context. If ServiceName is empty, all known properties are shown.
- Use the "ControlDataSource" to specify a data source suggesting selection values. This can be a selection list or a selection dialog that is opened to select a value.
- The "ControlType" specifies which type of control will be used (e.g. text, date or list control).
- If "ShowSecondControl" is set to "true", the control is shown as a "From To" field.
- The "LanguageKey" is used to set the control label.



Numerous changes in controls are only shown, when you save, close and restart the application!



Once you have created an item, e.g. input field, you cannot completely delete this item using the graphic configuration because the framework used does not support this function. To definitely delete an item that you have accidentally created, refer to section "1.5.1 Deleting items from detail applications".

1.2.5 Data sources

Data sources provide data for detail applications. This connection to a data source can be used to read data or to create, change or delete data. The repository defines for each data source the type of access provided to specified data.

Each application must include at least one data source, but can also include any number of data sources.

Adding/editing data sources

Use the button *Configure data sources* in the toolbar to create a new data source. Already existing data sources are edited via the same button. The dialogs used and the procedure are almost identical for both activities. It is therefore only described here how to create a new data source. You can easily use the description to edit an existing data source, too.

The "Configure data sources" dialog shows the list of the already existing data sources. If no data source was created, the list is empty.

By clicking on "Add", a new data source may now be created. Another dialog to specify the properties of the new data source is now opened¹. The following properties are available:

Data logic

The data logic specifies where the data source gets data. For this purpose, so-called data logics serving as connection to web services exist. The list shows for each data logic the name and a short description.

For each data source, you must specify exactly one data logic. The application designer must know which data logic provides the data required or performs the required functionality using the data (create, edit or delete).

Events

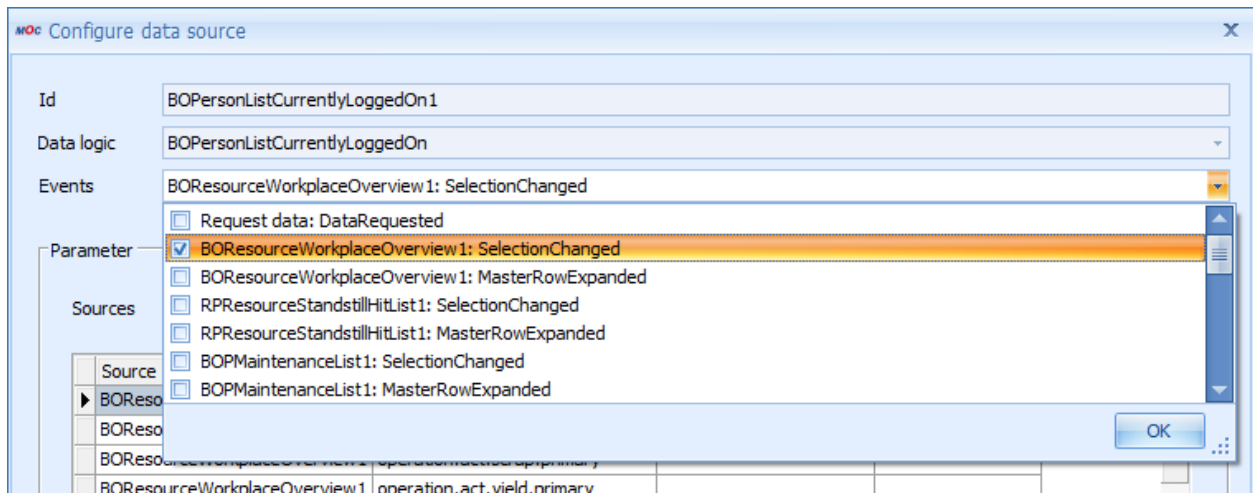
This field specifies the event. The data source then responds to this event. If this event occurs, the data source requests data from its data logic.

By default, only the event "Request data: DataRequested" can be selected. This means that the data source requests data when the green arrow in the toolbar is clicked.

In addition, each data source can respond to the SelectionChanged event of all other data sources created until then. This event occurs if one or more data records are selected in this data source (e.g. if you select a row in the annexed grid). This way, you integrate a master detail connection between the separate data sources. You often use this method to show detailed information for a row selected in the grid, e.g. all operations assigned to an order.

The list of available events is extended with each new data source. If a data source for the data logic WorkplaceOverview was created before, the selection looks as follows:

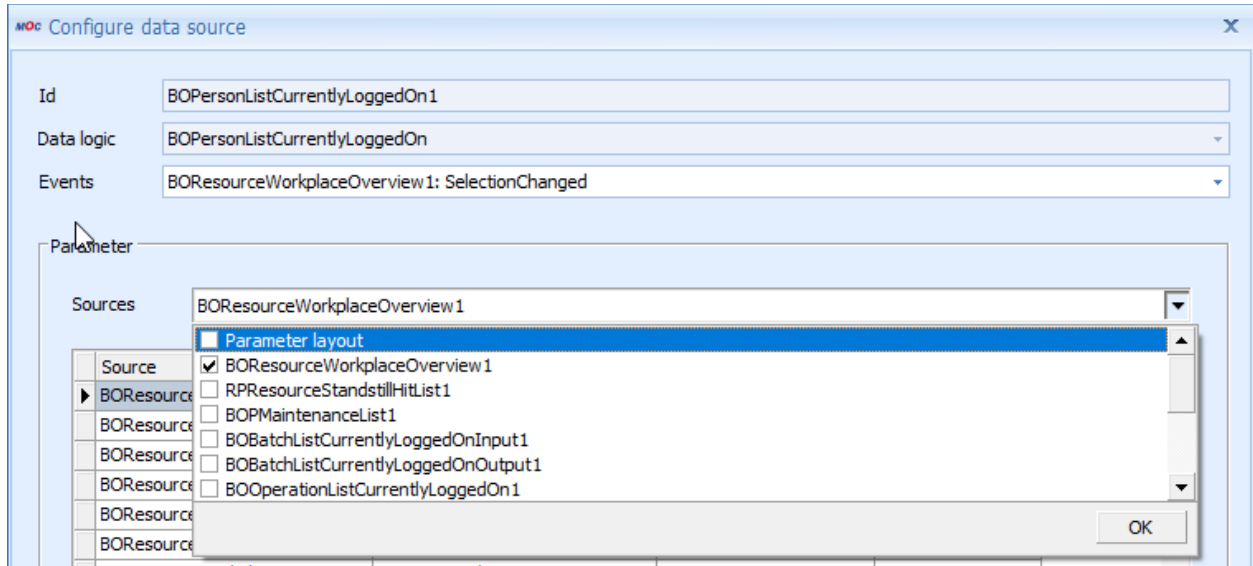
¹ By clicking on "Change", an already existing data source may be edited. A dialog is opened – the same dialog is used to create a data source. The already defined properties of the data source are pre-allocated.



Parameters: Sources²

You can parameterize the connection to the database that is created via data source. Using parameters, you can limit the requested data quantity or define the values of a new data record. The precise use of parameters depends on the type of data source.

The parameter sources specify where the data source gets the parameters. Each data source can get parameters from any number of sources.



If an application only includes one data source, you can only select the selection panel "Parameter Layout" as source. This means that the selection panel provides a parameter from the data source for each field (input field, checkbox, combo box, ...) of the selection panel.

² The parameter source is also called ParameterContainer.

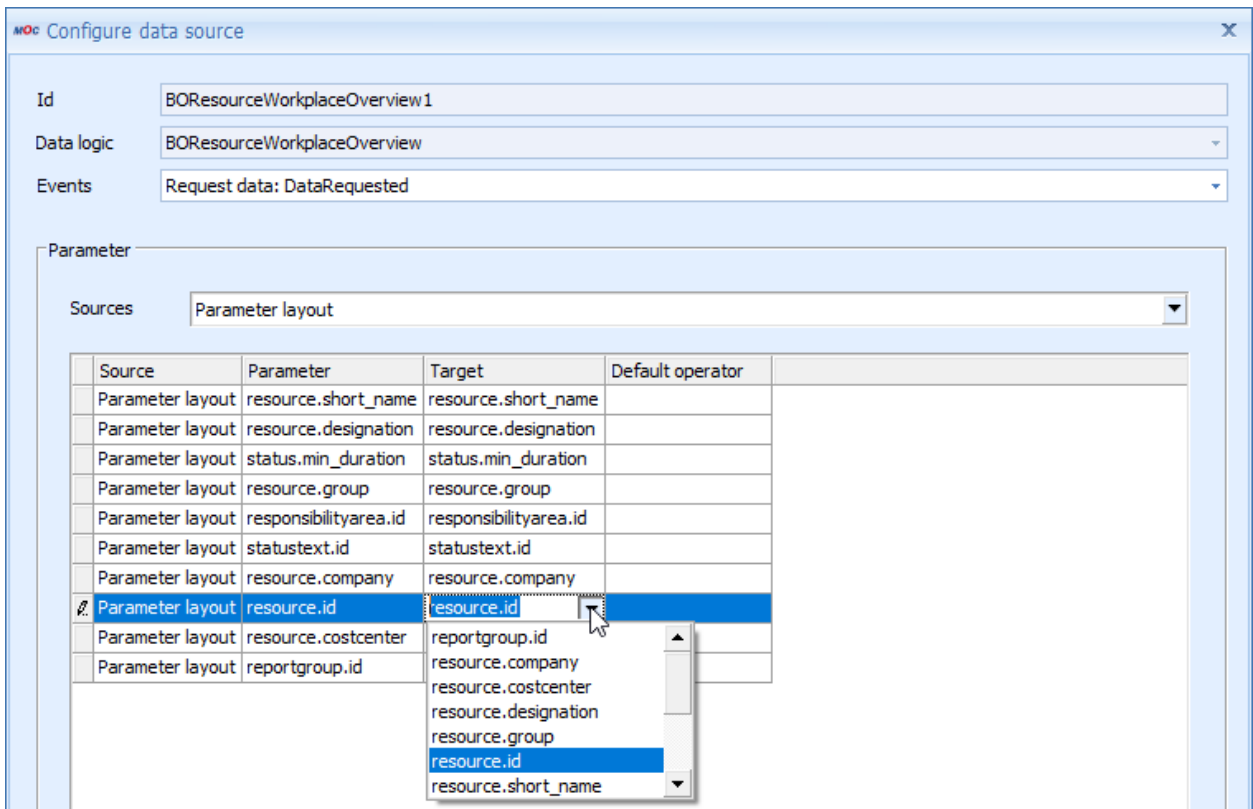
If several data sources are available, each data source can also obtain its parameters from the selected data records of all other data sources created until then. In this case, a parameter is created for each column of the dataset. The list of available parameter sources is extended with each new data source.

Parameters: Mappings

The parameters that a data source gets from different sources are not transferred to the data logic as they are. The user can define which parameters are assigned to parameters that are expected by the data logic. The user can also define how the parameters are mapped. An example can best illustrate the procedure.

Example 1:

In the selection panel, there is a field for the selection criterion *Workplace* with acronym "resource.id". The selection panel was selected as parameter source of the new data source.



Source	Parameter	Target	Default operator
Parameter layout	resource.short_name	resource.short_name	
Parameter layout	resource.designation	resource.designation	
Parameter layout	status.min_duration	status.min_duration	
Parameter layout	resource.group	resource.group	
Parameter layout	responsibilityarea.id	responsibilityarea.id	
Parameter layout	statustext.id	statustext.id	
Parameter layout	resource.company	resource.company	
Parameter layout	resource.id	resource.id	
Parameter layout	resource.costcenter	reportgroup.id	
Parameter layout	reportgroup.id	resource.company	

In the left column you see where the parameters are from. In this case, it is the selection panel. The second column shows how the parameters are identified within the selection panel. In the third column, a list of all parameters expected from the web service is shown. Now the requested target parameter may be selected for each parameter from the selection panel.

It is possible to map up to two parameters on the same target parameter. This will automatically be interpreted as from...to relation, e.g. to limit time ranges. At present you are not allowed to allocate the same target parameter to more than two parameters. In this case, the relevant parameters are not used in a data request.

The parameters of the selection panel can be allocated freely when fields are created. Note: due to the mapping, names of source and target parameters need not be identical. But the property `FieldName` of fields of the selection panel is used as key to identify specific properties like input screen, field width, label, etc. Carefully select the `FieldName` for this reason.

Example 2:

A data source for the WorkplaceOverview data logic has already been created and selected as parameter source for the new data source.

Source	Parameter	Target	Default operator
BOResourceWorkplaceOverview1	resource.group		
BOResourceWorkplaceOverview1	resource.id	bookingrelation.workplace	
BOResourceWorkplaceOverview1	resource.predicted_downtime		
BOResourceWorkplaceOverview1	resource.remaining_downtime		

Here, too, the two first columns show where the parameters are from and how they are called. In this case, one parameter, each, is offered for each field provided by the WorkplaceOverview data source. Since these fields, however, should normally not be included in the requests of the new data source as parameters completely, only those also allocated to a target parameter are considered.

Typical case in practice: WorkplaceOverview is used as data source for a grid. The new data source requires a machine number as parameter to identify all operations logged on to a specific machine, for example. In this case it is not reasonable to transfer all available parameters. Only the machine number (`resource.id`) is therefore mapped on the target parameter.

Note: In many cases, the acronyms of the source and target parameters are identical. This is often the case, but not always.

You can enter the operator in the fourth column that is used to map input to service parameter. In the example, the definition remains empty; this is the default operator "equals". Other options that are used less often are "like", "in", "lessThan", "greaterThan", etc.



If a parameter defined in the layout is not displayed as data source "Parameterlayout" in the list of available values, you must check if the relevant control has been properly configured. For example, if the data type of the parameter does not support an array (e.g. datatype "boolean" or .NET type "duration") and if the control has been defined using "ShowSecondControl=True", then the parameter is not included in the selection list because this assignment is not valid.

Cache time

This is used to specify the time in seconds which is added to the data logic when data is requested. Within this period, new requests with identical parameters are served from the cache. This value will overwrite a default value for the entire MOC.

Update rate

Time in seconds after which this data source will automatically and regularly request data. If the value is 0, the default value of the total application (see above) is used. If data is requested manually, the time is reset.

Column configurator

The column configurator is used to limit the data requested. Each data logic usually provides a very large number of fields (sometimes up to 1,000) because it is of universal use. But not the total number of fields needs to be displayed in a detail application. For reasons of performance, it is not reasonable to request data for fields which are not required anyway, the column configurator is used to specify the columns where data is actually requested.

The column configurator may be maintained via an additional dialog for each data source. All fields provided by the selected data logic are listed.

When creating a new data source, all fields appropriately identified in the repository are selected by default. If the data source is saved and the column configurator is not opened, then this preset selection is automatically used. To change the selection, select the required fields and close the dialog via OK.

Default values

A default value may be defined for each parameter expected from the selected data logic. This value is automatically allocated to the relevant parameter if no value is supplied by any parameter source.

Default values are also maintained via an additional dialog. All parameters expected from the web service are listed here.

The default value may be edited as free text. It is important, however, that the expected data type of the parameter is observed. The following values are possible:

Data type	Possible values
string	Deliberate character strings
integer	Only numerical values
decimal	Numerical floating point values
boolean	0, 1, false, true
datetime	Valid date character strings, e.g. 2009-02-01 or 02-01-2009.

If the "Array allowed" column includes the value "true", then a list of values can also be transferred to the parameter. You can enter a list of values; separate the values via semicolons (;).

If an invalid value is entered, this is notified to the user. The value will not be adopted in this case.

ResultSet

There are also data sources which may return several ResultSets. In this case, there are the following possibilities for customization:

- The data source is used by a specialized detail application plug-in capable of handling the supplied data structure (e.g. machine time profile, PDV evaluations/reports).
- In a detail application that can only process simple ResultSets, the first ResultSet is automatically displayed (important: the selection of the first set can be a selection by chance!).
- To specify the ResultSet that the DataController provides to "simple" detail applications, you can specify the name of the ResultSet to be used in the customization file. Currently, you can only do this by manually editing the file. You can find the name of the ResultSet in the repository column with the same name.

```

<DataControllerDescriptor>
  <Id>RPPProductionReportingStatusAnalysisByClassification</Id>
  <DataLogic>RPPProductionReportingStatusAnalysisByClassification</DataLogic>
  <ParameterContainers>
    <Mappings>
    <DefaultValues>
    <ColumnConfigurator>
    <CacheTime>0</CacheTime>

```

```
<RefreshRate>0</RefreshRate>  
<UpdateParametersByService>>false</UpdateParametersByService>  
<ResultSet>profile</ResultSet>  
</DataControllerDescriptor>
```

Important: If the result for the data source is specified, the same ResultSet is made available for all detail applications using this data source. If a specific detail application requires a different ResultSet, you must specify this via script.

Deleting data source

To delete a data source, the requested data source must be selected in the list of all data sources already created.

Click *Remove* to remove this data source from the application. For security reasons, the user must confirm the operation.



You can only delete a data source that is not used yet.

You can use a data source as

- Event "provider" for other data sources;
- Parameter source for other data sources;
- Data source for one or more detail applications.

In this example, all three conditions are true. The data source is actually used and you must check again if you really want to delete the data source. If yes, you must first remove all connection to the data source. With the data source "PersonListCurrentlyLoggedInOn2, you must remove the data source from the list of parameter sources and events. The "Workplace overview" detail application must also be deleted because no other data source can be selected for it subsequently.

1.2.6 Detail applications

Detail applications are used to present data in an application. There are different types of detail applications. Some of them are very universal and are therefore used quite frequently, e.g. grid, chart, detail display or pivot grid. Some of them refer to a special application case, e.g. cycle progression chart.

When you create a detail application, you specify its type, i.e. how data is presented. This specification is fundamental and cannot be changed once a detail application has been created.

Creating / editing detail applications

To create a new detail application, click the toolbar button *Configure detail applications*. Already existing detail applications are edited via the same button. The activities of creating and editing dialogs are very similar and the same dialogs are used. Below it is therefore only described how to create a new detail application. You can easily use the description to edit an existing data application, too.

The dialog *Configure detail applications* opens and a list of the already existing detail applications is shown. If no detail application has been created before, the list is empty.

Click *Add* to create a new detail application. Another dialog to specify the following properties is now opened.

ID

The ID must be clear within the application because the ID is used to clearly identify the detail application.

Caption

The text entered specifies the title of the detail application. The detail application is displayed within a dockable panel. The text entered here is displayed in the title bar of the panel.

Note: enter a language key here to guarantee the correct translation when changing to another language.

Please note: When the detail application has been created, the translation of the language key will directly be displayed in the list of existing detail applications. Requirement: the language key entered must already exist.

Application category

The application category specifies the form used to display the data. There are general application categories used for numerous different detail applications (chart, table view,...). There are also very special application categories that are only used for very special detail applications. It is possible that an application category is specified for one single detail application only. In this case it is possible that the data source, which provides the data, is specified in a fixed manner for the detail application. Example: the cycle progression. If the cycle progression is added to an application, the assigned data source ResourceCycleProgression with default values is automatically created, too. You can edit the data source later on when the application has been created.

It is also possible that the data source has already been created. In this case, the data source is not created again, but the already existing one is automatically assigned to the detail application.



In case of already existing detail applications, the application category cannot be changed subsequently.

The cycle progression, which includes its own data source, is a special case. This application category does not only include a chart for the display of data, but also additional control elements to limit data (e.g. radio buttons to select shifts/hours). This detail application therefore does not only provide its own data source but also its own parameters and its own events. The user expects that the data displayed is adapted to the selection upon a click on a radio button.

Events

Like data sources, some detail applications can respond to events. This always depends on the application category. If you select a relevant application category, this is possible. The selection list is then activated and the user can select all events where the detail applications of this type can respond.

Important difference to events with data sources: a data source always responds to an event in the same way, i.e. data is requested. With detail applications, the response is not clearly specified, but depends on the category of the detail application. For information on the response of a detail application to an event, refer to the documentation of the relevant detail application (see **Error! Reference source not found.**).

Data source

If the data source has not been specified automatically with the selection of an application category, you can select a data source here. All data sources are available, which have been created as described in 0. Select the relevant data sources. The new detail application will then get its data from the selected data source.



In case of already existing detail applications, you cannot change the data source subsequently.

Ribbon page

The ribbon page entered here is activated as soon as the detail application is focused.

Additional data sources

Some detail applications support the use of several data sources. If you select such an application category, this selection list automatically becomes active.

Authorization key

Authorization keys to control authorizations that are required to display this detail application. If no authorization key is defined, the detail application will always be opened.

Help file and Help index

Name of file where help subjects for this detail application are listed and/or description of the entry point within this file.

Icon

Selection of the icon displayed in the title bar of the detail application. Select the icon using an image selection dialog. After having selected an icon, the icon is displayed on the button that opens the dialog.

Show selected data records only

Usually, all data records of the selected data source are displayed in a detail application. In some cases this is not reasonable. Example: the application category "Detail view" is normally used to display specific values of a single data record in different fields.

Typical application case: several workplaces are shown in a table view. Next to the table and for reasons of overview, you want to show the ID, status, group, cost center, etc. of a single workplace in a detail view. The detail view must always provide details on the workplace selected in the table.

Here, both detail applications show data of the WorkplaceOverview data source, either all data or only one data record.

The same data source is assigned to both detail applications. But in the detail view, you only want to show the selected data record. You therefore enable the option *Show selected data records only*. All data records highlighted in the grid are selected. If several data records are highlighted, the data of the data record highlighted last is displayed in the detail view.

Column configurator

Use the column configurator to narrow down the data available in the detail application. In an ideal case, the relevant data source has already been configured accordingly and provides only the fields required. But some applications are very special and cannot "handle" all fields of their data source (e.g. specific charts).

You can edit the column configurator via an additional dialog for each detail application. All fields provided by the selected data source are listed (see screenshot 0).

When you create a new detail application, all fields are selected by default. If the detail application is saved then without opening the column configurator dialog, this pre-setting is automatically adopted. To change the selection, select the required fields and close the dialog via OK.



If no data source has been selected yet, this list is empty because the fields displayed are based on the data source.



If an additional column is added to the column configurator, a field is not automatically added to the detail application!

Deleting detail applications

To delete a detail application, select the required detail application in the list of all detail applications already created.

Click *Remove* to completely delete this detail application from the application. For security reasons, the user must confirm the operation.



You cannot use a detail application as a data source. In general, you can always delete a detail application. But if a data source has been created automatically for a detail application, this data source is not automatically deleted, too. If it is no longer required, you must delete the data source manually (see 0).

Showing detail application

When you have created a detail application, it is automatically loaded into a dockable panel and displayed in the application. You can place and resize the panel using drag and drop.

The detail application is automatically "initialized". For example, this can mean that in a grid, all columns are automatically added and shown that are available in the data source of the application.

Configuring contents of the detail applications

You are free to configure the contents of the different detail applications. With very simple detail applications like the type *Layout*, the fields displayed are configured like the selection panel (see 1.2.4). The configuration of more complex detail applications is described in the following sections.

1.3 Customization of editing applications

Editing applications are very similar to main applications. Their structures are nearly identical and they provide almost the same configuration options.

This section describes the customization options provided by editing applications. Many customization options are identical to those provided by main applications. This section therefore focuses on the differences or additional possibilities.

1.3.1 Application generator

Editing applications are usually created along with the relevant main application in an automated manner using the application generator.

For a detailed description of the procedure, refer to the document [MDS-ApplicationGenerator.pdf](#).

1.3.2 Additional customization options

In general, editing applications provide the same configuration options that are described above. Some further settings are only available for editing applications and are only active if an editing application is called.

When you create the application, some reasonable values are preassigned to these additional settings.

Currently, you can only perform a subsequent customization if you edit the main customization file of the application (<NameOfEditingApplication>.config in the application folder of the editing application).

ParentController

Name of the relevant data source of the higher level main application.

The application generator initially sets this setting automatically. To make subsequent changes, you must manually change the above-mentioned customization file.

Subject to the `ParameterProcessingMode`, the values from the currently selected row of this data source are transferred to the editing application and assigned to its fields.

Subject to `UpdateMode/UpdateSourceMode`, the data of this data source is updated after performing the editing function.

ParameterProcessingMode

Specifies if and how parameters from the calling main dialog are transferred to the editing dialog.

The application generator initially sets this setting automatically. To make subsequent changes, you must manually change the above-mentioned customization file.

The following options are available:

- `NoParameter`: no parameters are transferred.
- `SingleParameter`: transfers the selected grid row (mandatory); an error message is displayed if no row is selected.
- `OptionalSingleParameter`: transfers the selected grid row (optional); no error message.
- `MultiParameter`: transfers one or several grid rows (mandatory); an error message is displayed if no row is selected.
- `OptionalMultiParameter`: transfers one or several grid rows (optional); no error message.

- **SingleSelectionParameter:** transfers the selected grid row. If no row is selected, the values of the selection panel are passed. If the selection panel neither transfers values --> an error message is displayed.
- **OptionalSingleSelectionParameter:** transfers the selected grid row. If no row is selected, the values of the selection panel are passed. No error message.
- **SelectionOnlyParameter:** transfers the values from the selection panel. If no values are transferred → error message.
- **OptionalSelectionOnlyParameter:** transfers the values from the selection panel. No error message.

UpdateMode

Specifies how data is changed via editing function.

The application generator initially sets this setting automatically. To make subsequent changes, you must manually change the above-mentioned customization file.

The following options are available:

- Insert: new data records are added
- Update: existing data is changed / edited
- Delete: existing data is deleted

UpdateSourceMode

Specifies how data is updated in the main application, once the editing function has been executed.

You can make this setting initially via the application generator. To make subsequent changes, you must manually change the above-mentioned customization file.

The following options are available:

- All: all data of the main application is completely refreshed (using the values entered in the selection panel). This is required, for example, for separate delete and copy dialogs. Because here, you cannot identify the relevant data records and only request these data records.
- OnReturnValues: the service called returns values that are used to request only the relevant data. This may be the case, e.g. for "Insert". In this context, it is often the case that a new key field is generated that is required to request the new data record.

Important: if the service called does not return any fields that can be used to clearly identify a relevant data record, then this setting does not make sense. This setting is set by default for *Insert* and must be changed, if necessary.
- OnKeyValues: the key fields of the list service are used to request only the relevant data. This is the ideal procedure for the *Update*.

Important: the service of the main application and the editing service must have the same key fields; otherwise it does not work.
- OnScript: for future use

1.3.3 Process configuration

To configure the processes for editing functions as precisely as possible, a process configuration exists for each editing application in the file `ProcessConfiguration.config`.

The application generator automatically generates this file when the application is created. The file is directly stored in the application directory of the application.

Another section of this document describes the process configuration in detail.

1.4 Calling external programs

You can call external programs from the MOC via the menu or from an entry in the toolbar of an application.

1.4.1 Menu

To call an external program from the menu, you make an entry in the menu editor:

The following values are relevant:

- Enter "StartExternalProcess" as command and ID
 - Enter the program to be run in "Parameter"
- For further details, refer to section "Parameters"

Command	Labeling	ID	Command	Parameter	Im
Test Notepad	Test Notepad	StartExternalProcess	StartExternalProcess	path="notepad" example.txt	

1.4.2 Toolbar

To start an external program from an application, you can call the program via toolbar

You use the link editor for customization.

- Function: callCommandObject
 - Parameter
Program to be executed including parameters
- For further details, refer to section "Parameters"

ID	Call external Program
Command	StartProcess
Function	callCommandObject
Parameter	StartExternalProcess path="notepad" example.txt
Authorization key	
Visibility	
Title	lkEditExampleTxt
Tooltip	lkEditExampleTxt
Category	
Ribbon page	lkMainPage
Context	

1.4.3 Quick launch bar

Directly enter CommandLink to call the program

Example:

StartExternalProcess path="notepad" example.txt

1.4.4 Parameter

To call external programs, the following parameters are supported

Path

This parameter defines the program that is called

-> The "path" parameter **must** be specified.

-> The transferred value has to be enclosed by double inverted commas.

The specified program must be executable and/or if the program cannot be called automatically, the program including the entire program path must be entered

All other parameters are optional and transferred to the executing program as parameters.

Parameter

Variables can be transferred in square brackets.

List of parameters

These parameters are possible:

UserName:

logged on user

MesInstanceName:

registered system

MesInstanceUrl:

Url of the registered system

Language:

Language selected by the user (format: en-US)

All Application Settings

If you put [CONF:] in front, you can transfer any ApplicationSettings in the call parameters,

e.g.:

CONF:LookAndFeelSkin

the skin selected by the user

The complete list of possible application settings can be shown if you enter the transaction code "syssettings" on the MOC

Examples:

Starting the MaintenanceManager for the system logged on:

StartExternalProcess path=" iexplore.exe" "http://[MesInstanceUrl]/MaintenanceManager/"

Starting any program and transfer of skin

StartExternalProcess path="irgendeine.exe" Skin=[CONF: LookAndFeelSkin]

1.5 Tips & Tricks

1.5.1 Deleting items from detail applications

Background

You can use the context menu of the graphic configuration to delete items, e.g. input fields. But the framework used does not definitely delete the items. It only hides the items and they still exist as "hidden item". The framework does not provide a technical option to definitely delete an item created for a detail application.

Solution

If it is not enough to hide the items and if the items created must be deleted definitely, you must manually change the configuration files. To do so, you must delete XML entries in the layout configurations of the detail applications using a text editor or an XML editor.

Layout files are files with the extension "*.config" that include the text "layout" in the name. These files are stored in the directory of the application configuration of the MOC in the relevant scope. Example:

- d:\Moc\local\conf\Moc\Apps\Units\LayoutPanel.config
- d:\Moc\local\conf\Moc\Apps\Units\LayoutMainDetail.config
- d:\Moc\local\conf\Moc\Apps\Units\MDUnitsInsert\ParameterLayoutMDUnitsInsert.config

In the relevant XML file, search for the "FieldName" of the item that you want to delete (in our example "example.item.to.delete") and delete an XML node at two places.



When you edit XML files, the MOC must not be started. The changes are enabled with the next start of the MOC.

Deleting the item

```
<property name="Text"></property>
<property name="CustomizationFormText">SI-Einheit</property>
<property name="StartNewLine">false</property>
<property name="Visibility">Always</property>
<property name="TextLocation">Left</property>
</property>
<property name="Item7" isnull="true" iskey="true">
  <property name="TypeName">LayoutControlItem</property>
  <property name="ControlName">example.item.to.delete</property>
  <property name="AllowHtmlStringInCaption">false</property>
  <property name="TextAlignMode">UseParentOptions</property>
  <property name="SizeConstraintsType">Custom</property>
  <property name="Image" isnull="true" />
  <property name="ImageIndex">-1</property>
  <property name="ImageAlignment">MiddleLeft</property>
  <property name="ImageToTextDistance">5</property>
  <property name="OptionsPrint" isnull="true" iskey="true">
    <property name="TextToControlDistance">-1</property>
    <property name="AllowPrint">true</property>
  </property>
  ...
  <property name="TextLocation">Default</property>
</property>
...
```

Delete the complete item property (in the example Item7) from the opening to the closing tag (underlined).



The items contain numbers (e.g. "Item7") The numbers need not be consecutive numbers. It is therefore not necessary to assign new numbers to the succeeding items when an item has been deleted.

Deleting the settings of an item

```

<Setting Key="mpdvEdit example.item.to.delete" Description="" LastChanged="2017-07-
27T08:16:55.8650007Z" Value="<Value>
  <ConfigurationItem xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns="http://www.mpdv.de">
    <Acronym>example.item.to.delete</Acronym>
    <ServiceName />
    <Label>lkTest</Label>
    <Length>0</Length>
    <FillChar xsi:nil="true" />
    <ShowInGrid xsi:nil="true" />
...
    <CanNotEqualOrNull xsi:nil="true" />
    <TransferEmptyValuesToHydra xsi:nil="true" />
    <IsResult xsi:nil="true" />
    <ShowSecondControlInSearch xsi:nil="true" />
    <LoadDataOnInit xsi:nil="true" />
  </ConfigurationItem>
</Value>
</Setting>

```

Delete the complete setting from the opening to the closing tag (underlined).